GW — lab notebook (gravitational-wave program)

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Data of record: GWTC-2.1/3 PE (GWOSC/Zenodo, 76 events); GWTC-4.0 PE (Zenodo 16053484, 86 events); LVK TGR products (Zenodo 17461225: imr/par/rin/liv); GWTC-3 cosmology icarogw (Zenodo 5645777); PTA chains: NANOGrav 15yr fig-1 release, EPTA DR2 (Zenodo 8091568), PPTA DR3 (github). Every battery locked-preregistered before running; every headline independently re-derived; numbers of record in `results/`, per-battery detail in `reports/`.

Black-hole posterior geometry: the event atlas and its laws (E38, E40–E44)

E38 GW posterior-geometry atlas. Value/shape geometry on real compact-binary posteriors: all 76 confident GWTC-1/2.1/3 events (1.54M samples; GWOSC/Zenodo PE releases), 6 black-hole planes each — (m_1, m_2), $(\mathcal{M}, q)$, (q, χ_{eff}), (M_f, a_f), $(d_L, \cos\iota)$, (z, M_{tot}); 74/76 produce all six. The foundation for E40–E47.

E40 GW chirp-mass lawfulness. Is the (m_1, m_2) posterior degeneracy just the constant-chirp-mass line? Predict each event's orientation from the constant-$\mathcal{M}$ tangent at its median masses, vs a constant-total-mass null, 75 events. PASS: median $|\Delta\psi_{\text{chirp}}| = 7.49°$ vs total-mass 20.38°; chirp wins 61/75; on the elongated subset (axis ratio ≥ 3, n=28) 4.68°; Spearman(error, axis-ratio)$= -0.42$ (more elongated $\rightarrow$ tighter, as predicted). A rare POSITIVE geometric-lawfulness result — geometry is lawful where a dominant physical combination (chirp mass) exists.

E41 GW plane lawfulness map. A-priori correlation-sign predictions for all 6 planes, derived by 12 physicist/verifier agents blind to the data, tested against measured within-event ρ (75 events). 3/6 HIT:

E42 BH population + precision lawfulness. Precision test: chirp-mass fractional precision $\sigma_{\mathcal{M}}/\mathcal{M}$ vs SNR (75 events). PASS: $r = -0.50$ (p=$7{\times}10^{-7}$), OLS slope -1.22 (Fisher -1), and partial correlation with $\log\mathcal{M}$ at fixed SNR = +0.94 (the cycle-count law $N \propto \mathcal{M}^{-5/3}$). Posterior SIZE is lawful — completing orientation (E40) + correlation (E41) + size (E42). Population map: χ_{eff} mean +0.076, 36% median-negative, m_1 peaks near 10 and 35 $M_\odot$, 7 IMBH-scale remnants, 1 BNS; the geometric outliers are the physically extreme events (GW190814 q=0.11, GW191219 q=0.04 NSBH, GW200115 NSBH, GW190425 BNS).

E43 χ_{eff} per-event negatives. Does any event securely have $\chi_{\text{eff}} < 0$ (dynamical-formation signature)? From the full posteriors (75 events): 0 events with $P(\chi_{\text{eff}} < 0) > 0.90$ (GW191109 sits at exactly 0.900, 90% CI $[-0.60, +0.12]$ spans 0), 0 with a 95% CI below 0. NULL: E42's 36% median-negative is largely measurement scatter, not per-event-secure negatives.

E44 χ_{eff} hierarchical. Does the POPULATION require support below 0? Gaussian fit (flat and induced-prior-reweighted, 75 events) + a positive-truncation comparison. $\mu \approx 0.06$–0.09, $\sigma \approx 0.045$–0.061, negative fraction median $\sim 9\%$ — but a $\chi_{\text{eff}} \geq 0$ truncated population fits as well or better ($\Delta \ln L = -1.0/-0.74$), so the negative tail is a Gaussian-tail artifact, NOT required. Reinforces E43. (Does not overturn the LVK negative tail: a simplified pipeline — Gaussian-only, approximate prior, no selection function — simply does not independently demand it.)

Testing general relativity with black holes (E45–E47, E55, E57–E60)

E45 IMR-consistency test. LVK GWTC-3 Tests-of-GR: the inspiral- vs post-inspiral-inferred final mass/spin, deviation $(\Delta M_f/M_f, \Delta a_f/a_f)$, GR = (0, 0), 18 events. GR UPHELD — D_1 combined deviation $(-0.035, +0.002)$, GR at credible level 0.671; D_2 17/18 consistent (outlier GW190814 has an absent post-inspiral, a known systematic). D_3 (the framework's added test): the mean-deviation directions are COHERENT (Rayleigh p = 0.001, 13/18 with positive final-spin deviation), only partly along the shared degeneracy — a coherent sub-threshold waveform systematic that per-event tests miss.

E46 parameterized PN deviations. Fractional PN inspiral-phase deviations $\delta\hat{\varphi}_i$ (GR = 0), 9 events × 10 coefficients (SEOBNRv4). GR NOT violated. The cross-event sign-coherence test isolates `dchi0` (leading 0PN):

9/9 events positive, Bonferroni $p = 0.004$, combined $+0.109\,[0.040, 0.177]$ excluding 0 — the documented low-PN WAVEFORM SYSTEMATIC of a single-waveform analysis, not new physics. The non-coherent combined exclusions (`dchi3/5l/6l`, only 2–4/9 positive) are naive-product artifacts the coherence test correctly discounts. Single waveform; multi-waveform robustness (not run) would confirm the systematic origin.

E47 ringdown no-hair test. Black-hole spectroscopy: fractional deviation of the 221 overtone frequency/damping $(\delta f, \delta\tau)$ from the Kerr prediction, GR = $(0, 0)$, 22 events. GR/Kerr UPHELD — D_1 combined 0.75σ from GR ($Q = 0.246$, via a robust Gaussian product); D_2 22/22 consistent; D_3 ISOTROPIC (Rayleigh $p = 0.79$), no coherent deviation (unlike E45/E46 — the coherent systematics are inspiral-specific, absent in the overtone). A first-pass histogram-density product spuriously reported a 99.9% "no-hair violation" that contradicted D_2/D_3 (an event at Q=0.987 was only 0.25σ from GR); the Mahalanobis cross-check caught the binning artifact and the Gaussian product corrects it.

E55 multi-waveform robustness. E40's chirp-mass orientation across two independent waveform families (IMRPhenomXPHM vs SEOBNRv4PHM, 59 events): PASS — the law holds for both (median $|\Delta\psi_{\rm chirp}|$ $9.1°/6.1°$) and the orientations agree to 3.3° (elongated). Real geometry is waveform-robust; E46's low-PN `dchi0` coherence, by contrast, is a known waveform systematic (would not survive). DATA-BLOCK (honest): a direct same-event multi-waveform test of the E45/E46 GR-test coherences is impossible with GWTC-3 (the parameterized set is FTI/SEOBNR-only; TIGER/IMRPhenom was delayed) — would need the GWTC-2 (different events).

E57 black-hole area theorem. Hawking's 2nd law from GWTC-3: $A_{\rm final} \geq A_{\rm initial}$ using the inspiral progenitors (initial areas) vs the ringdown remnant (final area), disjoint/ independent (Isi 2021 setup). Only 4 loud events carry a ringdown remnant. HELD: no violations, area increases 5–79%, GW170104 strong single-event confirmation 0.92, stacked 0.92σ. Honest: crude independent-pairing + broad inspiral spin priors give GW150914 P=0.74, below Isi's 0.97 – a consistency check, not a stringent bound.

E58 massive graviton / Lorentz-violation. GW-propagation gravity test: modified dispersion $E^2 = p^2c^2 + A\,p^\alpha$ over 12 events × 8 orders (α=0 = graviton mass). GR UPHELD: no Lorentz-violation/dispersion signal in any event or order. Graviton mass (per-event, robust): best single-event $m_g < 6.7 \times 10^{-24}$ eV/c^2, median 1.8×10^{-23} (LVK combined 1.27×10^{-23}). Honesty catch: a naive posterior-product combined bound gave an absurd 5.9×10^{-27} (3 orders too tight, grid-unstable) — LVK use a flat-in-A prior and the prior samples are empty in the release, so it cannot be divided out; the per-event bounds are reported and the LVK combined quoted, not a fabricated number.

E59 ringdown mode consistency. Does adding the 221 overtone shift the remnant (M_f, a_f) from the 220-fundamental value? rin.zip Kerr_220 vs plain Kerr_221, 22 events. A naive test "fails" (7 events $> 2\sigma$, max 6.1), BUT the shift is COHERENT — M_f decreases in 22/22 (100%) and a_f in 21/22 (95%) with the overtone. A uniform all-event down-shift is the known ringdown- overtone SYSTEMATIC (start-time/modeling; the GW150914 overtone controversy), NOT a no-hair violation. The coherence lens (as in E45/E46) prevents a false violation claim; the clean no-hair result remains E47 (221 deviation params ~ 0, isotropic).

E60 model-independent ringdown no-hair. The strongest-in-principle ringdown test: the MEASURED 220 QNM frequency (damped-sinusoid fit, no Kerr assumption) vs the Kerr prediction $f_{220}(M_f, a_f)$ from the INSPIRAL remnant (Berti fit) — independent data. Kerr CONSISTENT: no event deviates $> 2\sigma$ (max $|z|$=0.85), combined 0.34σ. GW150914 clean: Kerr 236.6 vs measured 246.8 Hz (+5%). Weak (only 3 events have both a DS ringdown and an inspiral remnant; GW170814's DS frequency is broad/poorly constrained), GW150914 carries the confirmation. Ringdown no-hair trilogy: E47 (deviation params ~ 0) + E59 (remnant shift = systematic) + E60 (frequency matches Kerr) all uphold the Kerr nature.

Deriving the mass-plane law; the curved chirp-mass law (E65, E67)

E65 the E40 law derived — locked Fisher-rotation REFUTED; the posterior IS the constant-$\mathcal{M}_c$ curve. Locked hypothesis: E40's residual pattern (heavy events fail toward constant-$M_{\rm tot}$) is a Fisher-mixture rotation — inspiral cycles measure $\mathcal{M}_c$, the merger band measures $M_{\rm tot}$, so orientation should rotate with detector-frame mass. Model: two-direction Fisher mixture, analytic aLIGO PSD, one constant κ fit ONLY on the 14 locked E40-prereg events, scored out-of-sample on the other 61. OUTCOME: D1 marginal pass (7.45 vs

7.72°), D2 FAIL, D3 (parameter-free sign test) FAIL decisively — measured ψ does NOT sit between the chirp and total tangents; rotation fractions are mostly NEGATIVE (light elongated events overshoot PAST the chirp tangent, away from total), a sign no rotation model can produce. What the failure revealed (post-hoc, labelled): the posterior lies along the constant-$\mathcal{M}_c$ CURVE; the principal axis of a curved arc is its chord, not the median-point tangent. Zero-free-parameter curved law (uses only the $\mathcal{M}_c$ median + q marginal): median error 4.85 vs 7.49° (all 75), **0.81** vs 4.68° (elongated, 6×); GW190728's 14° "violation" reproduced to 1°. On-record GWTC-4 out-of-sample prediction: $< 2°$ for elongated O4a events. 5 contract tests; transient errno-60 read handled by retry; headlines re-derived by hand.

E67 GWTC-4 out-of-sample: the curved chirp-mass law PASSES at 1.26°**.** The E65 prediction — locked on record (commit 8d09cbf) from GWTC-1/2.1/3 data only, prereg pushed BEFORE any GWTC-4 file was opened — scored on the fully disjoint GWTC-4.0 O4a catalog (Zenodo 16053484; 86 events, all processed, zero dropped, all files h5py-validated). **D1 PASS: median** $|\psi_{\text{curve}} - \psi_{\text{meas}}| = 1.26°$ **on the 19 elongated events (threshold** 2°**)** — 18 of 19 land at 0.2–3.3° while the tangent carries systematic 4–11° errors that the curvature term removes (GW231118_090602: 10.7 → 0.5; GW231020: 8.0 → 0.2; GW230529 mass-gap: 2.9 → 0.3). D2 PASS: curve beats tangent on all 86 (7.53 vs 9.36°). D3: the E40 tangent law itself replicates out-of-sample (86% wins vs 81%; Spearman −0.40 vs −0.42). The E65 finding is hereby PROMOTED from post-hoc to out-of-sample confirmed: the (m_1, m_2) posterior geometry of GW events is the constant-$\mathcal{M}_c$ curve (zero free parameters — each prediction uses only the event's own median source $\mathcal{M}_c$ and q marginal). Headlines re-derived independently + one event recomputed end-to-end by hand (exact match).

Gravitational-wave cosmology: the spectral-siren lever (E66)

E66 spectral-siren H0 obstruction — REFUTED at GWTC-3 precision. Radical claim (E61's obstruction logic on GW): the spectral-siren H0 program is blocked by the $\sim 33\,M_\odot$ mass-peak location μ_g acting as an uncalibrated ruler. Test: linear-response couplings on the REAL LVK GWTC-3 joint cosmology+population posterior (icarogw release, $n = 13{,}533$; power-law+ peak, restricted flat-ΛCDM). Locked stop-and-diagnose fired mid-run: the multivariate response $C_{AB}C_{BB}^{-1}$ (partial) disagreed with the mandatory decile cross-check (marginal) — an apples-to-oranges comparison in the implementation, caught exactly as the two-estimator rule intends; fixed to the marginal degeneracy track, after which all sanity gates pass. RESULT: a $4\,M_\odot$ coherent μ_g drift moves H_0 by only −0.51 km/s/Mpc ($< 1 \Rightarrow$ REFUTED; robust $-0.41/-0.32$ across SNR cuts) — an order of magnitude below the CMB-vs-SH0ES gap and inside the chain's own ±3.3 statistical error. What survives: μ_g IS the single largest systematic lever (mass group 0.38 vs rate group 0.22 km/s/Mpc per 1σ), so the obstruction becomes live when spectral-siren precision reaches ~ 1 km/s/Mpc (O5/XG) — then an independently pinned or z-dependent peak is mandatory. Conservative caveat: the restricted $H_0 \in [65, 77]$ prior compresses couplings (lower bounds), but the verdict does not hinge on it.

The nanohertz band: the GW background across pulsar-timing arrays (E68)

E68 cross-PTA GWB anatomy — pure degeneracy-sliding; γ marches with time span. First value/shape battery in the nanohertz band, on real validated chains: NANOGrav 15yr (HD-correlated), EPTA DR2full (HD), PPTA DR3 (CRN), + EPTA DR2new as robustness; CPTA has no public GWB chain. D1: all pairs mutually consistent (1.0–1.8σ), Gaussian-faithful (sliced-W ratios 0.98–1.01). D2 HEADLINE: the constraint geometry is UNIVERSAL ($\rho(A,\gamma) = -0.85..-0.95$, $\psi = 110$–116° — one measurement kernel) and the pairwise mean offsets lie **94.6–99.9%** along the common A–γ degeneracy axis (mean 0.98; locked threshold 2/3) ⇒ classified degeneracy-sliding: everything the PTAs disagree about lives along the direction a common power-law fit slides when read over different bands. D3: the SMBHB circular prediction $\gamma = 13/3$ is excluded $> 2\sigma$ by NANOGrav ALONE ($z = +3.03$; EPTA-full +0.29, PPTA +1.25) — the SH0ES pattern (E52): one experiment carries the narrative; no combined number fabricated (E58 lesson). POST-HOC (labelled): $\hat{\gamma}$ is STRICTLY MONOTONIC in observing span (10.3 → 24.7 yr: 2.79 → 3.25 → 3.87 → 4.21), including EPTA's internal DR2new-vs-DR2full split (1.89σ, same telescopes, offset also along the degeneracy) — the coherent signature of a spectrum that flattens at high frequency relative to a single power law (or a span-correlated systematic). Successor prediction on record: longer-span subsets of the same array should fit systematically steeper γ (preregisterable at IPTA DR3 / NG 20yr). 4 contract tests; every headline independently re-derived; all probe values match the publications.

GW data beyond black holes: the neutron-star EOS transfer (E63)

E63 neutron-star mass-radius geometry. First transfer of the value/shape framework out of cosmology, into the neutron-star equation of state, in the (M, R) plane. Real posteriors: NICER pulse-profile for the low-mass star J0030 (Riley 2019 X-PSI ST+PST; Miller 2019 2-oval/3-oval) and the high-mass star J0740 (Riley 2021; Miller 2021), plus GW170817 tidal Λ mapped to (M, R) via the quasi-universal C–Love relation. Column identifications and every headline independently re-derived; 5 contract tests pass. METHODOLOGICAL LESSON: in (M, R) the axes are incommensurable ($M_\odot$ vs km), so the degeneracy ANGLE ψ is metric-dependent and meaningless (joint-whitening forces orthogonal long axes — a spurious ψ-gap$= 90°$ on every pair until caught); the scale-free shape invariant is the correlation $\rho(M, R)$. RESULT D1: the NICER pipeline/model systematic is VALUE-dominated — every Riley-vs-Miller and spot-model difference is a central-value offset at fixed orientation (orientation-only value fraction 0.90–1.00, $\Delta\rho \leq 0.08$); the sharpest is J0740, a +1.75 km radius offset (Miller 14.05 vs Riley 12.30) plus a 3.2× larger Miller ellipse. This is the OPPOSITE of the cosmology analog (the KiDS statistic-dependence, E48, was a precision/shape artifact). RESULT D3: $\rho(M, R)$ is organized by MASS REGIME and robust across independent pipelines — low-mass J0030 $\rho \approx +0.86$ (steep M–R banana) vs high-mass J0740 $\rho \approx +0.25$ (flat top of the M–R curve), regime $\Delta\rho = 0.61$, consistent across pipelines (0.84/0.89/0.86; 0.29/0.21) — the NS analog of "the kernel sets orientation" (organizer = location on the mass–radius curve). D4 (exploratory, flagged): X-ray J0030 vs GW170817 at $\sim 1.4\, M_\odot$ is MIXED — both a $\sim$1.5 km value offset (GW prefers smaller R, $\sim 1.1\sigma$) and $\rho : 0.86 \to 0.28$, but the GW ρ is C–Love-mapping-induced. D5: all six posteriors are near-pure ($\mathrm{Tr}\,\rho^2 = 0.990$–0.996); the $\sigma^2/8$ Chernoff repackaging (E62) persists. Honest limits carried up front: non-Gaussian (multimodal) NICER marginals, the mapping dependence of the GW point, and the QI repackaging.

GWTC-5/O4b: the curved chirp-mass law reconfirmed out-of-sample (E71)

E71 GWTC-5/O4b curved-law repeat. Third disjoint catalog after GWTC-1/2.1/3 (train) and GWTC-4/O4a (E67). Prereg locked (commit d19a7a8) BEFORE any O4b file was opened; decision rules byte-identical to E67, only a declared structural file/group adapter added. GWTC-5.0 O4b PE (Zenodo 20276106 + 20348006): 104 per-event releases, 52.84 GB, ZERO dropped, fully disjoint (event dates 240413–250119, zero name-overlap with the 86-event O4a set, 12 events from the 2025 tail). D1 (the on-record E65/E67 prediction; elongated axr≥ 3, $n = 32$): median $|\psi_{\mathrm{curve}} - \psi_{\mathrm{meas}}| = 1.22° < 2°$ — PASS (E67 gave 1.26°). D2: curve beats tangent, 5.41° vs 8.23°. D3: the E40 tangent law replicates a third time (median 8.23°; chirp-vs-total wins 88%; Spearman(err, axr) $= -0.69$, the strongest of the three catalogs). Zero free parameters — each prediction uses only the event's own median source M_c and q marginal. Waveform-robust: the curve tracks the measured axis WITHIN every waveform family to $< 0.7°$; the per-event largest-analysis pick (5 of 104 events do carry a Mixed group; the locked rule uses it) shifts only the absolute orientation. Verification four legs: 17/17 contract tests; independent numpy re-derivation of D1/D2; one event hand-recomputed from raw HDF5 via a different linear-algebra path (match to 0.004°); computed medians reproduce the release's published PESummaryTable masses (exact for most, to $< 0.6\%$ for a few late high-mass events), column identity confirmed catalog-wide. The mass-plane geometry law is now reproducible across THREE independent catalogs spanning the full GW history. POST-HOC (labelled): the largest curve residuals are all round posteriors (axr~ 1), the ill-defined-orientation regime the axr gate excludes — not law-breaking physics; the E72 outlier atlas must control for axr before reading residuals as anomalies.

E72 GW geometry outlier atlas. Prereg locked (0d1cebf) BEFORE any residual-axis correlation. Where does the constant-M_c curve law break? Searched the 32 elongated O4b events (axr≥ 3; residual r = err_curve from the locked E71 results) against 9 pre-declared axes with Benjamini–Hochberg FDR ($q = 0.05$) and partial-Spearman controlling axr; Tukey outlier fence; E59 coherence lens. RESULT — a clean PHYSICAL NULL: none of precession (χ_p), aligned spin (χ_{eff}), mass ratio, redshift, chirp/total mass, or secondary mass (NSBH/mass-gap proxy) survives FDR — the constant-M_c geometry has NO source-class-dependent breakdown in O4b (strengthens claim 1). The only surviving correlate is waveform-model disagreement (a systematic; $\rho = -0.64$, $q_{\mathrm{BH}} = 0.001$, robust to drop-one ≤ 0.036), and its NEGATIVE sign says not-physics: the events the curve fits best ($r \sim 0.01$–

0.2°) are those where waveforms disagree most — the curve tracks the waveform-ensemble consensus better than any single waveform tracks another (measurement optics; the E59 lesson again). One marginal Tukey outlier (GW240507, 3.37° vs fence 3.23°; low-SNR, asymmetric $q \sim 0.27$) logged as a WEAK preregisterable successor target, not a claim. Verify: 6 contract tests; the headline ρ and the Tukey fence independently re-derived (hand rank-Pearson; numpy percentile). Limits: $N = 32$ low power; loudness axis data-limited (6/32 picked groups carry a network-SNR column); eccentricity and higher-mode axes absent from the released PE (out of scope).

The loudest event: GW250114 no-hair deep-dive (E74)

E74 GW250114 deep-dive (loudest event). A single-event characterization + synthesis (NOT a locked battery): the O4b TGR products are not public, so the ringdown spectroscopy and no-hair verdict are the PUBLISHED result (Suzuki et al. 2026, arXiv:2605.03576, orthonormal QNMs; LVK IMR via GWTC-5). GW250114 is the loudest GW event to date (network SNR ~ 80), a near-EQUAL-mass BBH ($m_1 = 33.8$, $m_2 = 32.2\,M_\odot$, $q \sim 0.96$, low spin), remnant $M_f^{\rm src} = 62.8\,M_\odot$, $a_f = 0.679$ — my PE extraction reproduces the published IMR remnant exactly (the data gate). INDEPENDENT contribution: propagating the remnant posterior through the Berti–Cardoso–Will Kerr fits gives the no-hair QNM targets — $(2,2,0)$ at 248.9 Hz / $\tau = 4.10$ ms and the $(2,2,1)$ overtone at 243.4 Hz / $\tau = 1.35$ ms (tight to ~ 1 Hz, the remnant being measured to $\sim 1.5\%$). The published analysis detects the $(2,2,1)$ overtone at 99.9% (up from 82.5% non-orthogonal) and finds $\delta f_{221}, \delta\gamma_{221}$ consistent with zero — Kerr no-hair UPHELD at record SNR. GW250114 is essentially a louder GW150914-twin: near-identical remnant ($M_f \sim 68\,M_\odot$ detector, $a_f \sim 0.68$; GW150914-like $f_{220} = 250$ Hz) but $\sim 3\times$ the SNR, which is why the once-contested overtone is now a clean detection. Curved-law N/A (near-equal-mass $\Rightarrow$ round posterior, axr~ 2, outside the E71 elongated set). Verify: 4 QNM contract tests (GW150914 reference, $f \propto 1/M$, spin trend, overtone damps faster); published-value validation of all four remnant/mass quantities. Limit: the strain spectroscopy is not reproduced — the no-hair verdict is the cited published result; the area theorem awaits the O4b TGR release.

A geometric test of GR: the inspiral mass-combination exponent (E78)

E78 posterior-geometry test of GR (the inspiral exponent). EXPLORATORY/post-hoc, grown from E71/E72: since the (m_1, m_2) posterior orientation is a zero-parameter function of the constant-$\mathcal{M}_c$ curve, INVERT the law and measure which chirp-mass exponent the geometry prefers — a test of GR through posterior geometry, not waveform templates. Generalize the curve to $\mathcal{M}_p = (m_1 m_2)^p (m_1 + m_2)^{1-2p}$ (GR: $p = 3/5 = 0.600$, set by the 0PN phase $\propto \mathcal{M}_c^{-5/3}$); the orientation is SCALE-FREE ($m_1(q;p) \propto q^{-p}(1+q)^{2p-1}$), isolating p from the mass scale. On the 32 elongated O4b events: $\hat{p} = 0.616 \pm 0.011\,(\mathrm{stat}) \pm 0.004\,(\mathrm{waveform})$ vs GR 0.600 — CONSISTENT at 1.3σ, a $\sim 2\%$ geometry-only confirmation of the inspiral mass-combination exponent. Validated: injection recovers the true exponent unbiased (bias +0.0002; $0.55 \to 0.550$, $0.65 \to 0.651$); the waveform systematic is negligible (± 0.004 — the per-event $\sim 3°$ scatter is incoherent and averages out); 5 contract tests; sensitivity $\mathrm{d}\psi/\mathrm{d}p \sim 30°$/unit. A deviation would have signalled modified inspiral phasing (dipole radiation / varying-G); none seen. HONEST: a methodologically independent cross-check, NOT more sensitive than the LVK -1PN dipole bound; O4b-only; the $\sim 1.2°$ E71 law residual sets a per-event floor (~ 0.04 in p), so the small +0.016 offset is at the level of the law's own imperfection, not claimed as physics. Successor (preregisterable): the fitted exponent is reproducible across GWTC-3/O4a/O4b and = 0.600.

E79 cross-catalog exponent — a 3σ "deviation" that is a systematic. Tests E78's successor (prereg locked BEFORE O4a was opened): is the geometry-fit exponent reproducible across catalogs and = 0.600? On GWTC-4/O4a (86 events, disjoint from O4b): $\hat{p}(\mathrm{O4a}) = 0.647 \pm 0.014$ (3.3σ from GR), $\hat{p}(\mathrm{O4b}) = 0.616 \pm 0.011$; they REPRODUCE each other (D2 PASS) but combined $\hat{p} = 0.628 \pm 0.009$ sits a naive 3.1σ above 0.600 (D1 FAIL). The prereg pre-committed a follow-up before any claim — and it dissolves the anomaly: (i) the two independent catalogs disagree by $0.031 \approx$ the offset 0.028, so the bootstrap stat error underestimates — a systematic floor ± 0.016 gives 1.5σ; (ii) the offset is elongation-dependent (Spearman(p^*, axr) $= -0.41$, coherent in BOTH catalogs) and the cleanest, most-elongated half gives $\hat{p} = 0.606 \approx$ GR. By the coherence lens a cross-dataset-coherent anomaly is a SYSTEMATIC — here a finite-width / higher-order imperfection of the pure-0PN constant-$\mathcal{M}_c$ model, vanishing in the clean limit — NOT a GR violation. Honest headline: the effective exponent is $0.628 \pm 0.009\,(\text{stat}) \pm 0.016\,(\text{syst})$, GR-consistent at 1.5σ, cleanest events 0.606; the naive 3σ is a false alarm the discipline caught (the E47/E58 lesson, live). Verified: injection unbiased on both catalogs; disjointness = 0; column-ID clean; honest budget independently re-derived. Successor: a 1PN geometric model should reproduce the -0.4 elongation trend and flatten it to 0.6; add GWTC-3 as a third point.

The information anatomy of a detection (E73)

E73 information anatomy of a detection. CHARACTERIZATION (forward Fisher from the waveform model + PSD, NOT strain-extracted; the data-driven test is the successor). For all 190 real O4a+O4b events, the frequency by which 50%/90% of the Fisher information about each parameter is accumulated — chirp mass (0PN, $\sim f^{-5/3}$), mass ratio (1PN), spin (1.5PN). RESULT: the ordering $\mathcal{M}_c < \eta < \chi$ holds for ALL 190 events — chirp mass is always learned first (~ 35 Hz), spin last; the anatomy "richness" ($\chi_{f50} - \mathcal{M}_{c,f50}$) has Spearman $= -1.00$ with total mass (lighter / longer-inspiral $\Rightarrow$ wider anatomy; heavy events merge almost immediately in band and compress it). A frequency-resolved "when is each thing measured" anatomy of every detection. GR-test direction (a -1PN dipole term): the RANK is reliable (Spearman $+1.00$ with mass; the lightest events — GW230529, the mass-gap merger, and GW230518 — are most sensitive, matching the known low-mass preference of dipole tests), but the absolute forecast is conditioning-limited (Fisher cond $\sim 10^{12}$; arbitrary β units), so only the ordering is quoted. Verify: 5 contract tests ($\mathcal{M}_c < \eta < \chi$ ordering, $f \propto 1/M$, $\sigma_\beta \propto 1/\rho$, lighter$\rightarrow$tighter dipole); conditioning explicitly checked; ordering + richness independently re-derived; suite 37/37. Honest ceiling: model-PREDICTED, not strain-MEASURED — the data-driven test (re-analyse loud events in frequency slices, compare empirical vs GR-Fisher information accumulation) is the successor, along with a numerically-stable ppE dipole forecast versus the LVK -1PN bound.

The GW inference manifold: the catalog as one geometric object (E82)

E82 the GW inference manifold. CHARACTERIZATION (post-hoc; radical idea made concrete). Treat all 190 O4a+O4b (m_1, m_2) posteriors as ONE object — each a Gaussian $N(\mu, \Sigma)$, the catalog the cloud of them, distance = 2-Wasserstein (location $|\Delta\mu|^2$ + Bures shape). RESULT: (1) location(population) 95% vs shape(measurement) 5% — mass-range-dominated, partly trivial; (2) the curved law slaves each ellipse's orientation to its location (median 1.26°, E71; residual physics-blind, E72), so orientation is NOT an independent axis; (3) classical-MDS variance 97%/3%/0%, participation-ratio effective dimensionality = 1.07 (naive per-event DOF = 5); axis 1 = POPULATION ($|\rho_{\mathrm{Mc}}| = 0.99$), thin axis 2 = PRECISION ($|\rho_{\mathrm{axr}}| = 0.51$). VERDICT: the catalog of black-hole mergers, geometrically, is a population sequence (chirp mass) dressed with a thin, law-determined measurement structure — subtract the law and the population remains; measurement and population are geometrically separable, and here separated. The catalog-level form of the program's thesis (E71 orientation = law(location); E72 residual physics-blind): the measurement optics occupy, and REMOVE, exactly one geometric axis. Verify: 4 contract tests (Bures closed form vs scipy `sqrtm` to 10^{-8}; MDS recovers synthetic 1D/2D dimensionality); suite 41/41. Honest: the 95/5 split and eff-dim$\sim$1 are mass-range-dominated (not oversold); Gaussian approximation; mass plane only. Successor: the full-parameter-space (spin, distance) Fisher-Rao version — the deferred E80 sloppy-spectrum program.

A strain-level pipeline, validated on GW250114 (E83)

E83 from-scratch strain pipeline. CAPABILITY + consistency check ("science first"): the strain-level analysis the exotic-test class requires, built with numpy/scipy only (no gwpy). Download GWOSC O4b strain $\rightarrow$ whiten (off-source Welch PSD) $\rightarrow$ bandpass $\rightarrow$ instantaneous frequency (Hilbert) for the chirp; post-peak dominant-frequency + log-envelope damping for the ringdown. GW250114 (H1): whitened peak 388σ (loudest to date); the CHIRP is recovered (f_{inst} rising through merger 89 $\rightarrow$ 117 $\rightarrow$ 187 $\rightarrow$ 220 Hz across ± 5 ms) and the RINGDOWN is recovered $f_{\mathrm{dom}} = 241$ Hz (3.2% from E74's Kerr 248.9 Hz), $\tau = 4.7$ ms (E74 4.10) — CONSISTENT with the E74 Kerr prediction. This independently cross-checks E74: the strain ringdown matches the Kerr QNM computed from the INSPIRAL-derived remnant — inspiral$\rightarrow$remnant$\rightarrow$ringdown closes in the actual data. Verify: 4 contract tests (a 250 Hz/4 ms damped sinusoid injected into noise is recovered; whitening flattens red noise; the bandpass suppresses out-of-band tones $> 1000\times$; a chirp's instantaneous frequency rises); suite 45/45. HONEST: a consistency check, NOT a measurement of record — single-detector, single-mode, fixed-start point estimate (few-% frequency / $\sim$15% τ bias vs a proper Bayesian multi-mode ringdown). Successor: a Bayesian ringdown; the data-driven frequency-geometry test (E73 successor); the exotic echo / dark-matter-frequency-rotation tests ONLY after multi-event validation + coherence-lens/systematic controls (per E79/E72, a positive there is likelier a systematic than new physics).